

1 Introduction

Artificial intelligence (AI) is poised to transform the global economy by simultaneously affecting aggregate demand and productivity on the supply side. Yet robust evidence on its firm-level effects outside the United States (US) remains scarce. While optimistic projections predict that AI could drive productivity booms (Baily, Brynjolfsson, and Korinek, 2023), others caution that adoption barriers and productivity constraints from unbalanced growth may significantly limit the gains from AI adoption (Acemoglu, 2024; Filippucci, Gal, and Schief, 2024). Europe’s position is both central and paradoxical: it boasts world-leading researchers and strong industrial capacity but suffers from a persistent and growing gap in the development of new AI technologies compared to the US and China (EIB, 2025). Understanding how AI affects European firms is therefore essential for policymakers seeking to harness its benefits while managing the associated risks.

This paper provides the first causal evidence on how AI adoption affects productivity and employment in European firms, leveraging a unique survey on EU and US non-financial firms matched to administrative financial data. We address two core questions. First, we explore the firm characteristics that predict AI adoption in Europe, which is key to characterize the selection process into using AI tools. We then investigate how AI adoption influences firm productivity and employment. The first question thus provides the empirical groundwork for a credible identification of the second.

The analysis reveals four key insights. First, AI adoption increases labor productivity by 4% on average after addressing endogeneity. Second, the gains stem from capital deepening rather than job displacement. Third, benefits are concentrated in medium and large firms – posing risks of widening inequality in the benefits from AI adoption. Fourth, the productivity gains are larger if firms carry out complementary investments in software and data or employee training. Our findings cast some doubts on claims of dramatic AI-driven job destruction, while underscoring pressing concerns about its uneven economic impact.

We leverage data of the European Investment Bank Investment Survey (EIBIS), an annual survey covering approximately 12,000 non-financial firms across the EU and 800 US non-financial firms, matched with ORBIS data on corporate financial statements. Our final sample used in the analysis is a pooled cross-section that runs from 2019 to 2024. The survey’s stratified design ensures representative coverage across firm size, sector and